Year 12 Further Maths - Work and energy

Similar questions - Slopes, impulses and flow

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A crate of mass 6 kg is pulled from rest along rough horizontal ground by a force of 40 N acting at 30° above the horizontal. The coefficient of friction between crate and ground is 0.25. The crate moves 3 m.

(a) Find the normal reaction between the crate and the ground. **[2]**

(b) Use an energy method to show that the speed of the crate after 3 m is 4.99 $\mathrm{m\,s^{-1}}$ (3 s.f.). **[4]**

The force is then removed and the crate slides on to a smooth slope inclined at 25° to the horizontal.

(c) Find the distance the crate slides up the slope before it comes to instantaneous rest. **[3]**

2 A puck of mass 0.4 kg is at rest on a rough plane inclined at 15° to the horizontal. It is struck by an impulse of magnitude 2.1 N s directed up a line of greatest slope. The resistance to motion while the puck is moving is a constant 1.2 N.

(a) Find the initial speed of the puck. **[2]**

(b) Find how far up the slope the puck travels. **[4]**

(c) Given that the maximum friction is also 1.2 N, show that the puck stays at rest once it stops. **[2]**

(d) Suppose instead that the resistance is a constant 0.6 N. Find the speed with which the puck returns to its starting point. **[6]**

3 A pump raises water through a straight pipe of length 60 m, inclined at 20° to the horizontal. Water leaves the top of the pipe at a speed of 8 $\mathrm{m\,s^{-1}}$, and the pump delivers 15 kg of water each second. Assume that there is no resistance to the flow and that the water enters the pipe at rest.

(a) Find the gain in gravitational potential energy and in kinetic energy of 1 kg of water. **[3]**

(b) Find the rate at which the pump does work on the water. **[3]**

(c) The pump is 70% efficient. Find the power input to the pump. **[2]**

4 A particle B of mass 3 kg is at rest at the bottom of a rough slope inclined at 15° to the horizontal. A constant force D acts on B directly up the slope. The resistance to the motion of B is a constant 9 N. When D stops acting, B has moved 40 m up the slope and its speed is 12 $\mathrm{m\,s^{-1}}$.

(a) Find the work done by D, and hence the magnitude of D. **[4]**

(b) Find the time for which D acts, stating the assumption you make about the acceleration. **[3]**

(c) Find the average power of D over this time. **[2]**

Question 1

(a) $R = 6g - 40\sin 30^\circ = 58.8 - 20 = 38.8$ N

(b) $F = 0.25(38.8) = 9.7$ N. Work done by the pull $= 40\cos 30^\circ \times 3 = 103.9$; work against friction $= 29.1$. Net $= 74.82 = \frac{1}{2}(6)v^2$, so $v^2 = 24.94$ and $v = 4.99\ \mathrm{m\,s^{-1}}$

(c) $\frac{1}{2}(6)(24.94) = 6g\sin 25^\circ d$: $d = 24.94/(2g\sin 25^\circ) = 3.01$ m

Question 2

(a) $I = mv$: $v = 2.1/0.4 = 5.25\ \mathrm{m\,s^{-1}}$

(b) $\frac{1}{2}(0.4)(5.25)^2 = 5.5125 = (0.4g\sin 15^\circ + 1.2)d = 2.2146d$, so $d = 2.49$ m

(c) Component of weight down the slope $= 0.4g\sin 15^\circ = 1.01$ N, which is less than the maximum friction 1.2 N, so the puck does not slide back

(d) Up: $5.5125 = (1.0146 + 0.6)d$, $d = 3.414$ m. Down: $\frac{1}{2}(0.4)v^2 = (1.0146 - 0.6)(3.414) = 1.4155$, so $v = 2.66\ \mathrm{m\,s^{-1}}$

Question 3

(a) GPE $= g \times 60\sin 20^\circ = 201$ J per kg; KE $= \frac{1}{2}(8)^2 = 32$ J per kg

(b) Rate $= 15(201.1 + 32) = 3500$ W (3497)

(c) Input $= 3497/0.7 = 5000$ W (4995)

Question 4

(a) Work by $D = \Delta$KE + PE gain + work against resistance $= \frac{1}{2}(3)(144) + 3g\sin 15^\circ(40) + 9(40) = 216 + 304.4 + 360 =$ 880 J. Magnitude $= 880.4/40 = 22.0$ N

(b) Constant acceleration from rest: $40 = \frac{1}{2}(0 + 12)t$, so $t = 6.67$ s

(c) Average power $= 880.4/6.667 = 132$ W